HEALTHCARE ANALYTICS EXECUTIVE REPORT

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Hospital Operations, Cost and Patient Flow Intelligence Executive Report

Analysis Period: 2011 – 2022

Executive Summary

This analysis evaluated 11 years of healthcare activity, financial performance, operational efficiency, procedure utilization, and patient return behavior. The main findings are summarized below:

- 27,891 patient encounters and 47,701 procedures, generating approximately $101.5 million in claim costs.
- Uninsured patients accounted for 48.5% of total claim costs.
- Medicare contributed the largest share of payer coverage, accounting for 61.8% of total reimbursement received by the hospital.
- The average encounter duration was 7.3 hours, and the median duration was only 15 minutes, indicating that a relatively small number of long-duration encounters place disproportionate demands on hospital resources.

Overall, the findings indicate that the hospital's greatest opportunities for improvement lie in addressing the financial impact of uninsured patients, improving reimbursement performance, understanding the drivers of repeat patient utilization, and optimizing resource allocation for high-cost and long-duration encounters. Continued monitoring of these areas will be critical to supporting both financial sustainability and operational efficiency.

Business Objectives

The healthcare organization requested analysis of its historical data to understand its operations, encounter costs, payer coverage, procedure utilization, and patient return rates. The data contains information regarding encounter records, patient records, payer details, and organization records.

Management aims to understand:

- Encounter volumes and costs
- Procedure utilization
- Payer coverage
- Hospital Operations

This analysis will enable the service delivery team, operations manager, and financial manager make informed data-driven decisions regarding the hospital’s operations.

Key Findings

Healthcare Activity

Encounter volumes remained relatively stable over the 11-year reporting period, although a spike in encounter volume was observed in 2014.

- The hospital recorded 27,891 patient encounters over the 11-year reporting period, generating approximately $101.5 million in total claim costs. Payer contributions totaled just over $31 million, resulting in an overall coverage rate of 30.6%. This indicates that a significant proportion of healthcare costs were not covered by payers.
- Patients without insurance accounted for the largest share of claim costs, generating approximately $49 million, equivalent to 48.5% of total claims costs.
- Despite accruing approximately $4 million in claim costs, encounters associated with Humana members received no payer coverage. This anomaly warrants further investigation to determine whether it is the result of data quality issues, claim processing practices, or coverage limitations.
- Encounter volumes remained relatively stable throughout the reporting period; however, a notable increase occurred during January 2014. Ambulatory encounters accounted for 60.3% of this increase, while outpatient encounters contributed 17.9%. Further investigation is required to identify operational or external factors that may have contributed to this activity spike.

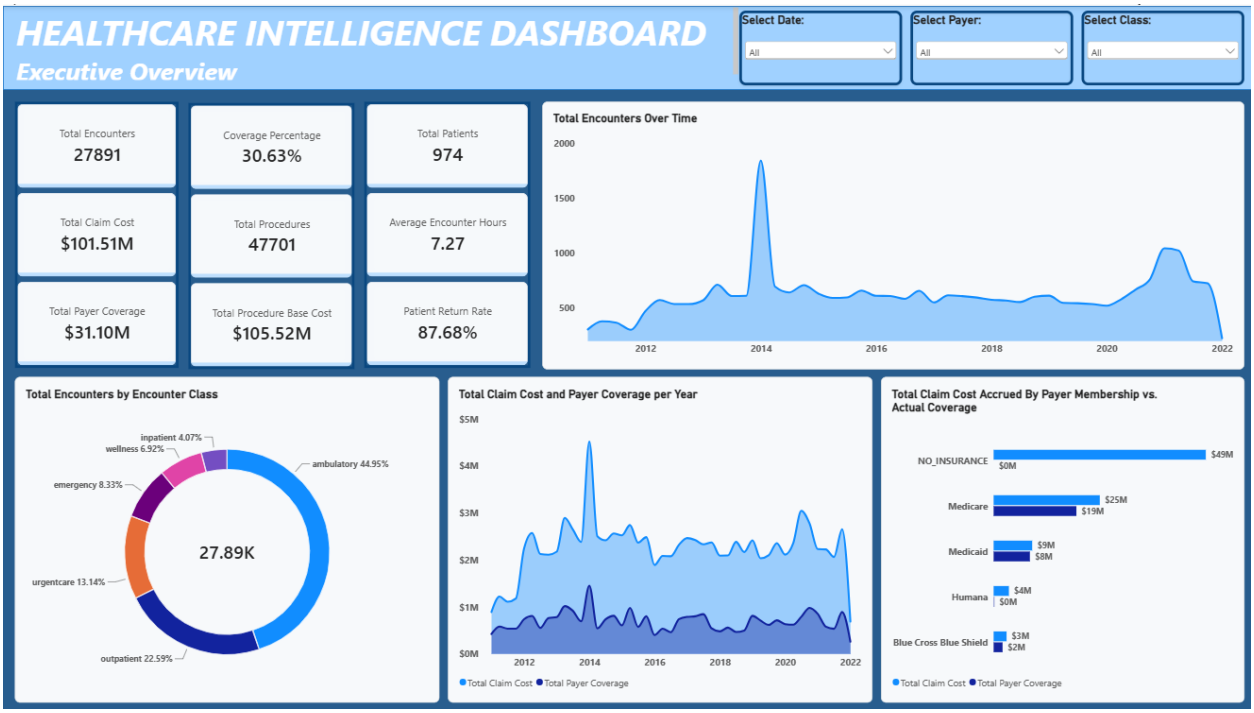


Figure 1: Executive Overview Dashboard.

Financial Performance

Analysis of insurer performance revealed substantial variation in average claim costs across payer groups. Medicaid members recorded the highest average claim costs, while uninsured encounters also generated significant healthcare expenditure. These findings indicate that healthcare costs and payer contributions vary considerably across patient groups.

- During the reporting period, the average encounter generated approximately $3.64 thousand in claim costs, while average payer coverage amounted to approximately $1.11 thousand per encounter.
- Ambulatory and urgent care encounters generated 58.7% of total claim costs. Average encounter costs were approximately $2.9 thousand for ambulatory encounters and $6.4 thousand for urgent care encounters. Inpatient encounters were the most expensive encounter type, averaging approximately $7.8 thousand per encounter.
- Payer coverage varied significantly across encounter classes. Payers covered approximately 36% of ambulatory encounter costs, 13% of urgent care costs, and 44% of inpatient encounter costs.
- Medicaid members recorded the highest average claim costs, accounting for 17.7% of total average claim costs. Patients without insurance accumulated an average hospital bill of approximately $5.6 thousand per encounter.
- Although Medicaid members incurred the highest average encounter costs, Medicare provided the greatest overall financial contribution, accounting for 61.8% of total payer coverage received by the hospital. Medicaid contributed 27.1% of total payer coverage.
- The relatively low overall payer coverage rate appears to be influenced by the substantial volume and cost of encounters associated with uninsured patients. Further investigation is recommended to identify the primary drivers of uncovered healthcare costs.

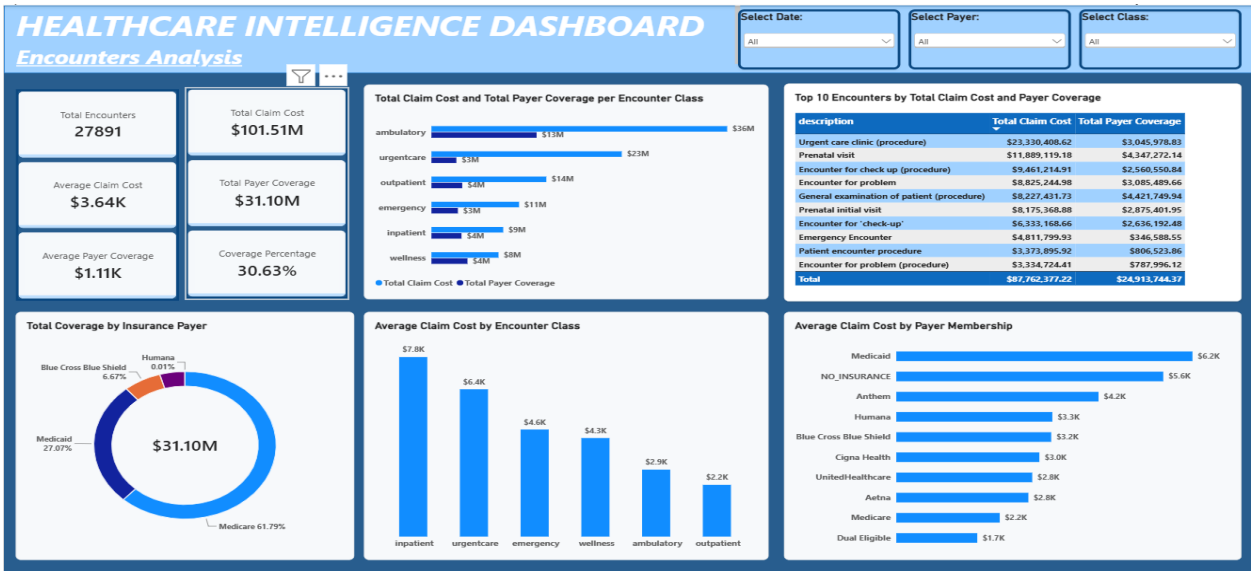


Figure 2: Encounter Analysis Dashboard.

Procedure Utilization

Procedure utilization analysis identified a small number of procedures that contributed disproportionately to overall healthcare activity and financial expenditure.

- A total of 47,701 procedures were performed during the reporting period. The associated procedure base costs amounted to approximately $105.5 million, reflecting the cost of keeping beds open and maintaining hospital capacity, staffing, and operational readiness.
- The hospital incurred an average procedure base cost of approximately $2.2 thousand per procedure.
- Despite the increase in encounter volumes observed during January 2014, procedure base costs remained relatively stable throughout the reporting period.
- Procedures averaged 9 minutes and 36 seconds in duration (0.16 hours), while the longest recorded procedure lasted approximately 18 hours.
- The most frequently performed procedure was the Assessment of Health and Social Care Needs procedure, with an average base cost of approximately $431.
- Further investigation is recommended to determine the relationship between procedure duration and procedure-related base costs.

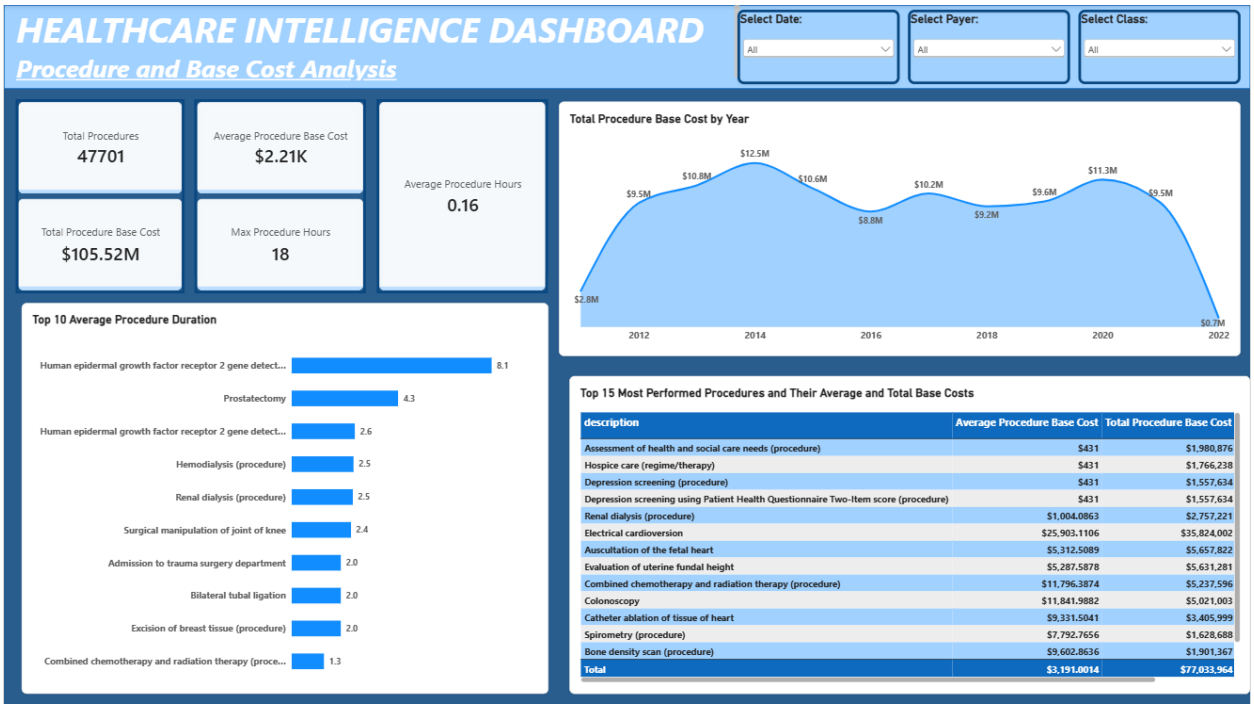


Figure 3: Procedure and Base Cost Analysis Dashboard.

Operational Efficiency

Operational analysis focused on encounter durations and resource utilization

- The average encounter duration was approximately 7.3 hours, while the median duration was only 0.25 hours (15 minutes). This substantial difference indicates a highly skewed distribution, where a relatively small number of long-duration encounters significantly increase the overall average.
- Total encounter base costs amounted to approximately $3.2 million, with the hospital spending an average of $116 per encounter to maintain healthcare capacity and service availability.
- Only 75 encounters exceeded 24 hours in duration, representing approximately 0.27% of all encounters. Inpatient encounters recorded the longest average duration at approximately 37 hours, suggesting that extended hospital stays were concentrated within inpatient services.
- Additional analysis should be conducted to identify the procedures associated with long-duration encounters and to understand the factors contributing to extended patient stays.
- Wednesdays and Thursdays recorded the highest average encounter durations, indicating increased operational activity on these days.
- Procedure activity was highest on Mondays, with approximately 8,000 procedures performed on average during the reporting period, making Monday the busiest procedural day.

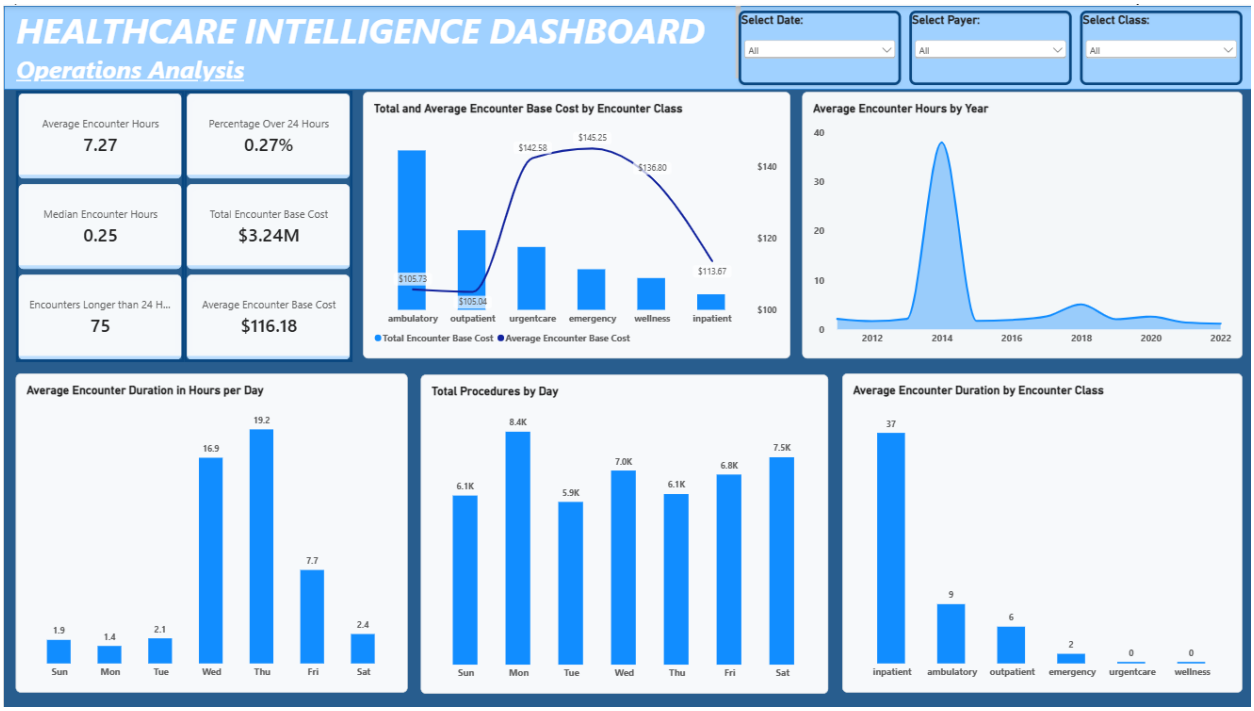


Figure 4: Procedure Utilization Analysis Dashboard.

Patient Return Behavior

Patient return patterns varied across encounter classes and payer groups, suggesting that certain healthcare services are associated with higher levels of repeat utilization. This information may assist healthcare providers with long-term planning, patient management initiatives, and service optimization.

- Patient return behavior remained relatively stable across the reporting period, with a noticeable increase observed in 2021.
- The hospital recorded a patient return rate of 87.7%, with 854 of 974 patients experiencing more than one encounter during the reporting period. This suggests a high level of repeat healthcare utilization and indicates the need for further analysis of the relationship between return behavior, encounter duration, and healthcare costs.
- Medicare, Medicaid, and uninsured patients recorded the highest number of returning patients. Humana patients recorded the lowest return rate at approximately 68.0%, despite limited payer coverage being observed for this group.
- Ambulatory and outpatient encounter classes generated the largest number of returning patients. Additionally, all patients associated with wellness encounters recorded repeat visits.
- These findings suggest that many repeat encounters may represent planned or ongoing treatment pathways rather than unplanned returns. Additional investigation is required to determine the underlying drivers of repeat utilization.

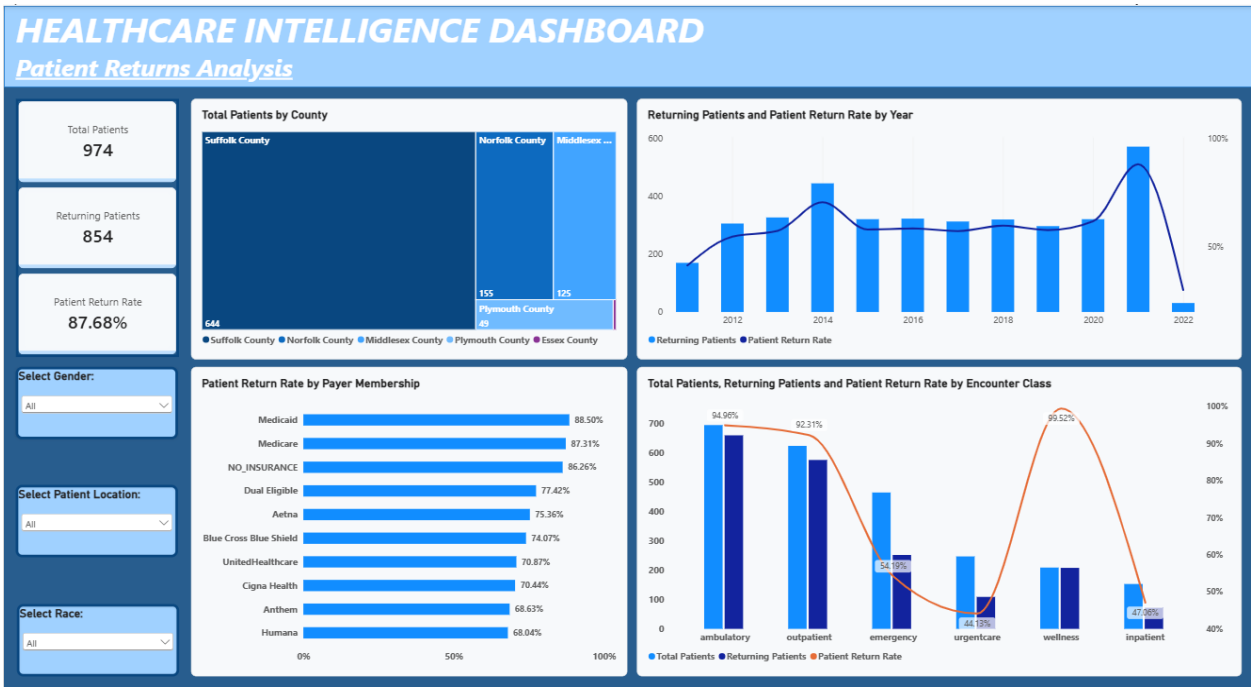


Figure 5: Patient Returns Analysis Dashboard.

Business Recommendations

Based on the findings generated by the analysis, the following recommendations are proposed:

1. Monitor High-Cost Healthcare Services

Healthcare administrators should monitor encounter classes associated with the highest claim costs and evaluate opportunities to improve efficiency without compromising the quality of care. Understanding the drivers of these costs may support more effective budgeting, resource allocation, and operational planning.

2. Review Uninsured Encounter Costs

Uninsured encounters contribute substantially to healthcare expenditure. Additional analysis should be performed to identify the factors driving these costs and to assess opportunities for improved reimbursement strategies or financial assistance programs.

3. Investigate Long-Duration Encounters

Although encounters lasting longer than 24 hours represent a small proportion of overall activity, they may place greater demand on healthcare resources. Further investigation should be conducted to determine whether these encounters are concentrated within specific encounter classes, organizations, or patient groups.

4. Improve Resource Allocation for Procedures

The organization should continue monitoring procedure frequency and cost trends to identify services that require additional staffing, equipment, or operational resources. High-cost procedures should be reviewed to evaluate opportunities for improving efficiency without compromising patient care.

5. Leverage Patient Return Insights

The high patient return rate presents an opportunity to better understand repeat healthcare utilization. Healthcare administrators should investigate whether return patterns are associated with chronic conditions, recurring treatments, or specific healthcare services. Understanding these patterns may assist with long-term capacity planning and patient management strategies.

Assumptions

Several assumptions were made during the analysis and dashboard development process:

- The source datasets accurately represent healthcare operations and patient activity during the reporting period.
- Each patient identifier represents a unique patient record.
- Each encounter identifier represents a unique healthcare encounter.
- Patients with more than one recorded encounter were classified as returning patients for the purpose of calculating the patient return rate.
- Encounter timestamps accurately reflect the duration of care received by patients.
- Procedure base costs reflect the operational costs associated with providing healthcare services and maintaining healthcare capacity.
- Payer coverage values accurately represent the amounts contributed by insurers toward healthcare claim costs.
- Encounter, procedure, and financial records were assumed to be complete for the reporting period following data quality validation.

Limitations

Although the dataset was assessed and validated prior to analysis, several limitations should be considered when interpreting the findings.

- The analysis is based on historical healthcare records and does not necessarily reflect current healthcare operations or payer arrangements.
- The datasets do not include clinical outcomes, patient satisfaction measures, staffing information, or resource availability data, limiting the scope of operational analysis.
- Some demographic and patient-related attributes contained missing values, reducing the depth of certain demographic analyses.
- The patient return rate measures repeat utilization rather than true patient retention, as the dataset does not provide sufficient information to calculate formal healthcare retention metrics across defined reporting periods.
- The reasons behind repeat encounters, prolonged encounter durations, and variations in payer coverage cannot be determined from the available data alone.

- Observed relationships within the data represent associations and trends and should not be interpreted as evidence of causation.

Risks

The analysis identified several areas that may represent potential operational or financial risks for the healthcare organization.

Financial Risk

Patients without insurance accounted for approximately 48.5% of total claim costs. This large concentration of uncovered healthcare expenditure may place financial pressure on the organization if reimbursement mechanisms are limited.

Coverage Risk

The overall payer coverage rate of 30.6% indicates that a substantial proportion of healthcare costs are not covered by insurers. Persistent gaps between healthcare costs and payer contributions may affect long-term financial sustainability.

Long-Duration Encounter Risk

Although only 0.27% of encounters exceeded 24 hours, these encounters may consume a disproportionate amount of healthcare resources, including staff time, bed occupancy, and operational capacity.

Patient Demand Risk

The high patient return rate of 87.7% may indicate recurring utilization patterns that place ongoing demand on healthcare resources. Without further investigation, it may be difficult to determine whether these return encounters are expected components of patient care pathways or indicators of inefficiencies.

Data Risk

Several data quality issues were identified during the project, including date formatting challenges and payer coverage import inconsistencies. While these issues were corrected, ongoing monitoring of data quality is necessary to maintain reliable reporting.

Opportunities

The analysis also revealed several opportunities for operational improvement and future investigation.

Improve Financial Sustainability

Further analysis of uninsured encounters may help identify opportunities to reduce uncovered costs, improve reimbursement rates, or implement targeted financial assistance strategies.

Enhance Resource Planning

Understanding the drivers of long-duration encounters may support more efficient allocation of beds, staff, and healthcare resources, ultimately improving operational efficiency.

Optimize High-Cost Services

Encounter classes and procedures associated with the highest costs should be reviewed to identify opportunities for process improvements, cost optimization, and resource utilization improvements.

Leverage Patient Return Behaviour

The high patient return rate presents an opportunity to better understand recurring healthcare needs, chronic condition management, and treatment pathways. Insights derived from repeat utilization patterns may support improved patient care and long-term planning.

Investigate the 2014 Utilization Spike

The significant increase in encounter activity observed during January 2014 presents an opportunity for deeper analysis. Understanding the factors that contributed to this increase may provide valuable operational and planning insights.

Expand Future Analytics

Future iterations of the dashboard could incorporate clinical outcomes, patient satisfaction data, staffing information, reimbursement performance, and predictive analytics to provide a more comprehensive healthcare performance management solution.

Conclusion

The analysis suggests that the hospital is delivering a high volume of healthcare services while operating under financial and operational pressures. Although more than 27,000 encounters and 47,000 procedures were performed during the reporting period, payer coverage accounted for only 30.6% of total claim costs. This suggests that a substantial portion of healthcare expenditure is either being covered by the hospital or remains unrecovered from the payers.

A major contributor to this coverage gap is the large volume of uninsured patients, who generated nearly half of all claim costs. This finding highlights a significant financial sustainability risk and

suggests that uninsured encounters should be a strategic focus for future financial planning and reimbursement management efforts.

Operationally, the hospital appears to function efficiently for the majority of encounters, with a median encounter duration of only 15 minutes. However, a relatively small number of extended encounters significantly increase average encounter duration and may place disproportionate demands on healthcare resources. Understanding the factors that drive these long-duration encounters may present opportunities to improve resource allocation and operational efficiency.

The unusually high patient return rate suggests that the hospital serves a population requiring ongoing and recurring care. While repeat encounters may reflect successful continuity of care and structured treatment pathways, they may also indicate the presence of chronic conditions or recurring healthcare needs that require further investigation. Understanding the drivers of repeat utilization will be important for future capacity planning and patient management strategies.

Overall, the findings indicate that the hospital's primary challenges are not related to healthcare demand, but rather to the financial burden associated with uninsured patients, low overall payer coverage, and the resource demands created by recurring patient utilization. Future strategic efforts should focus on improving payment recovery strategies, understanding the causes of repeat encounters, and optimizing the management of high-cost and long-duration healthcare services.